Are You Smarter than DoRA?

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Dora's Motivation

Can she solve interesting math problems using efficient fine-tuning?

Methods

LoRA: Low Rank Weight Update

$$W' = W_0 + BA$$

Key DoRA Insight: Separate Magnitude and Direction

$$W' = m\frac{W_0 + BA}{||W_0 + BA||_c}$$

0.71% of total trainable params! 0% inference latency!

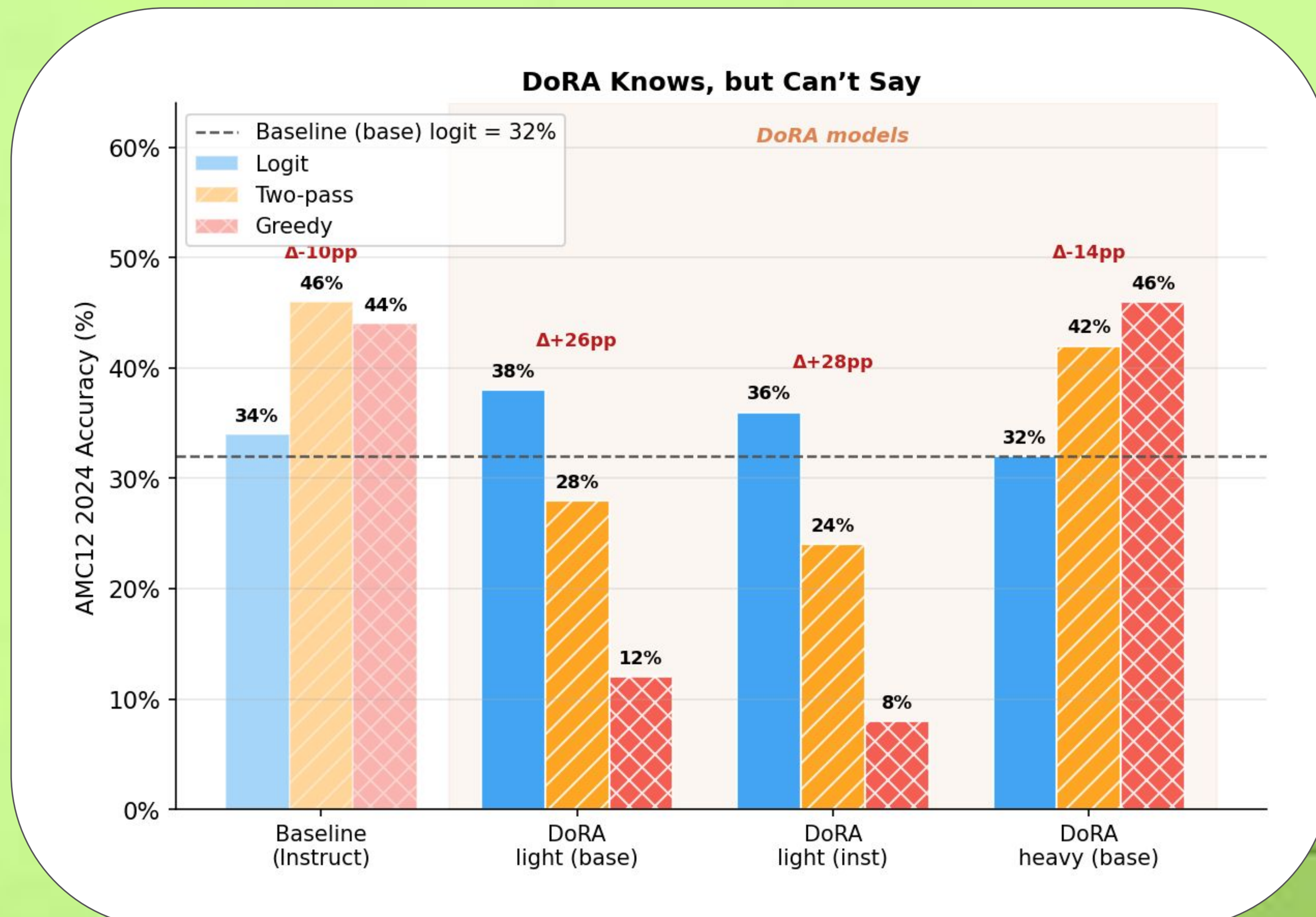
Model: Qwen 2.5-3B; Data: MATH-lighteval NuminaMath-CoT; Eval: AMC 12, 24/25

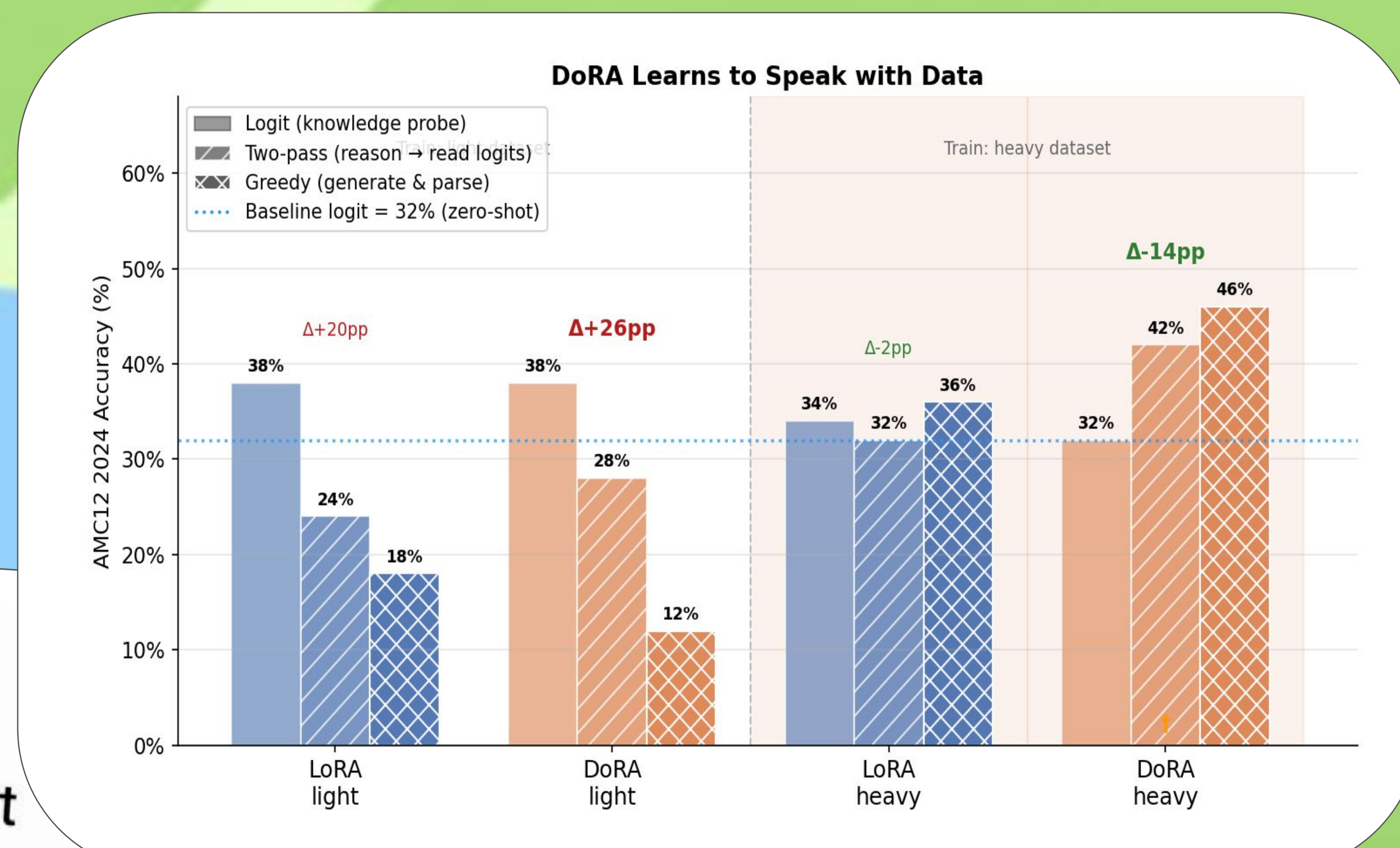
Problem

The product of three integers is 60. What is the least possible positive sum of the three integers?

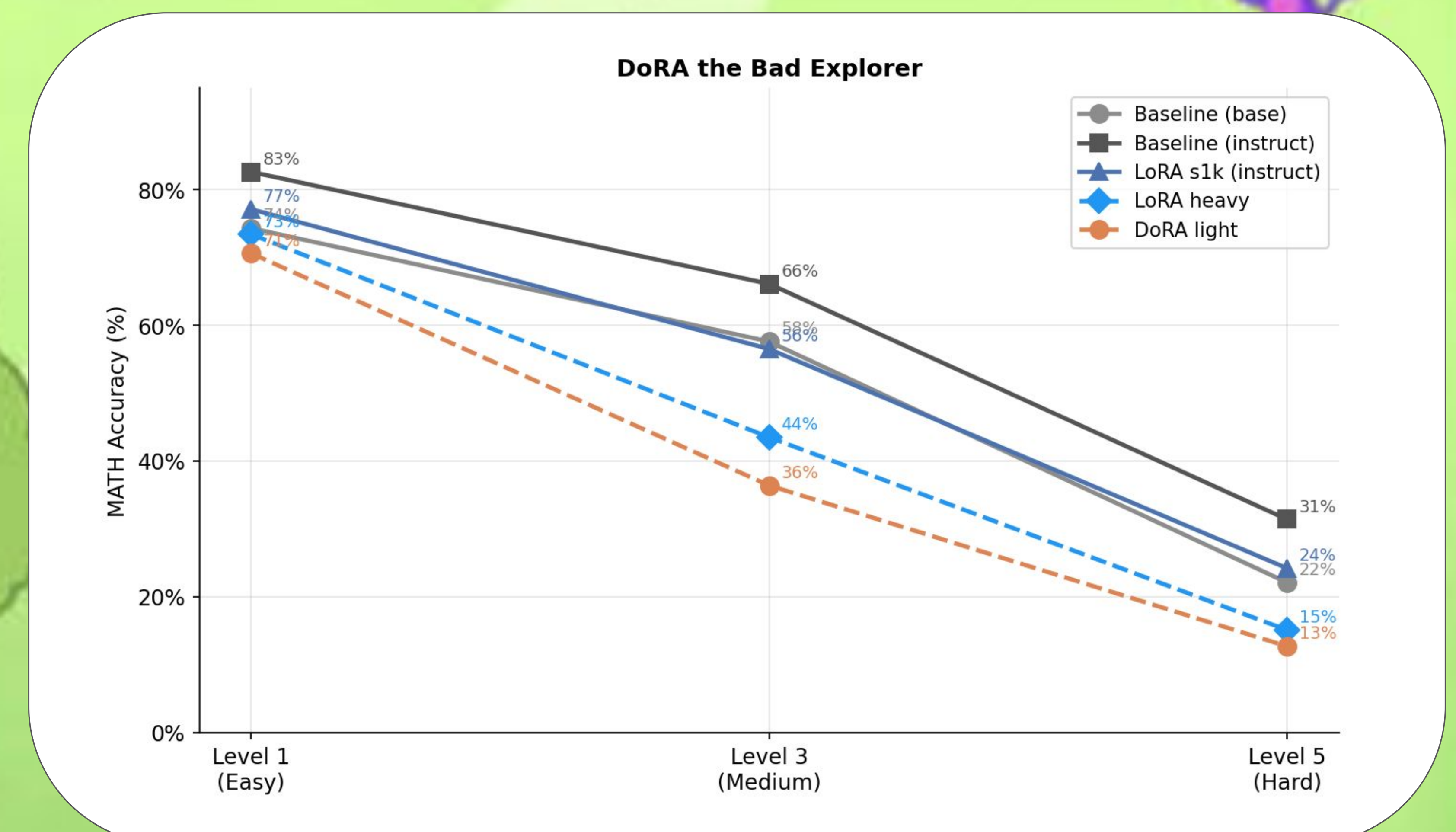
(A) 2 (B) 3 (C) 5 (D) 6 (E) 13

Results





Conclusion



Dora's Next Mission

- Harder tasks – AIME / Math Olympiad
- Alternative decoding interventions
- Bigger base model

Citations

[1] Liu, S.-Y., et al. *DoRA: Weight-Decomposed Low-Rank Adaptation.* ICML 2024.
[2] Hu, E. J., et al. *LoRA: Low-Rank Adaptation of Large Language Models.* ICLR 2022.
[3] DigitalLearningGmbH. *MATH-lighteval dataset,* Hugging Face, 2024.
[4] Li, J., et al. *NuminaMath-CoT dataset,* AI-MO, Hugging Face, 2024.
[5] Mangrulkar, S., et al. *PEFT: State-of-the-art Parameter-Efficient Fine-Tuning.* Hugging Face, 2022.